Task 5 — Steepest local search with a list of improving moves

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Solver revision: working tree on top of `81c7b03` (Task 5 strategies; this revision adds the 200-run experiment, the workbook check, the report generator and the move-by-move trace test)
Source code link: GitHub
Instances: `TSPA.csv`, `TSPB.csv` — 200 nodes each
Methods: `RandomSolution` (the starting solutions and the scale baseline), the full-neighbourhood steepest local search with a random start (the reference), and the same search driven by the list of improving moves (the new method); the two Task 4 candidate-list searches are included as previous methods for context
Previous methods: the Task 4 candidate searches, registered in the same benchmark run so their timings are comparable; the algorithms are unchanged

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1. The problem

The problem is the same as in Tasks 1–4. Each instance has `N` points with coordinates and a **cost**. We select exactly

```
M = ceil(N / 2) = 100
```

points and put them in a closed tour. The objective is

```
objective = tour length + cost of the selected nodes
```

and lower is better. Distances are Euclidean and rounded to the nearest integer, so all objective values are integers. A solution is stored as the list of selected nodes in visit order; the last node connects back to the first one.

Task 5 does not change the objective or the neighbourhood. It changes **how often the same delta is computed**. The full steepest search from Task 3 scans the whole neighbourhood in every iteration. Most moves are not affected by the move that was just accepted, so their deltas are still valid. The task is to keep those deltas in a **list of improving moves** (`LM`) and to use the list instead of scanning the neighbourhood again. The search should then reach the same quality in less time.

2. Methods

2.1 The list of improving moves

`LM` is a list of moves ordered from the best (most negative delta) to the worst. Only moves that improve the current solution are stored. The implementation uses a binary min-heap keyed by `delta`, so the best move

is popped first and insertion costs `O(log |LM|)`.

A move is stored by **node ids**, not by positions. This matters because a position in the tour is not stable: a 2-opt move reverses a whole segment, and an inter-route move replaces a node. The removed edges, on the other hand, are defined by their endpoints, so an entry that stores the endpoints stays valid after later moves.

```
move
    kind       inter or 2-opt
    delta      objective change at the time the move was evaluated
    a, b, c, d node ids (the meaning depends on kind)
```

The mechanism is separate from candidate moves. The LM search uses the **full** neighbourhood (every inter-route exchange and every 2-opt exchange), and the candidate lists from Task 4 are not used. The two mechanisms could be combined, but this experiment keeps them separate.

2.2 Inter-route moves

An inter-route move replaces one selected node `b` by one unselected node `d`. Let `a` be the predecessor of `b` and `c` its successor. The removed edges are `(a,b)` and `(b,c)`; the added edges are `(a,d)` and `(d,c)`. The delta includes the node costs:

```
Δ = cost[d] − cost[b]
  + dist[a][d] + dist[d][c]
  − dist[a][b] − dist[b][c]
```

The move is stored as `(a, b, c, d)`. One entry is enough: unlike a 2-opt move, a node replacement has no second reconnection, and the formula is symmetric in `a` and `c`. If the current neighbours of `b` are `a` and `c` in either order, the stored delta is still the exact delta of replacing `b` by `d`.

2.3 2-opt moves and the two edge orientations

A 2-opt move removes two edges and reconnects the four endpoints. For the edges `(a,b)` and `(c,d)` there are two ways to write the reconnection:

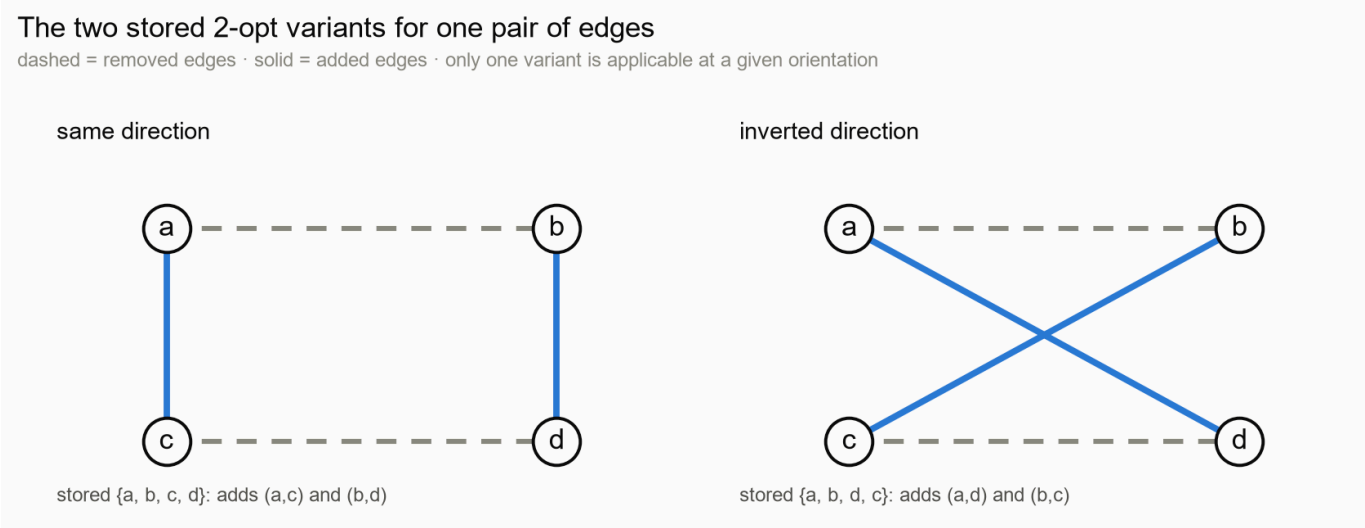
```
same direction:      add (a,c) and (b,d)
                     Δ  = dist[a][c] + dist[b][d] − dist[a][b] − dist[c][d]

inverted direction:  add (a,d) and (b,c)
                     Δ' = dist[a][d] + dist[b][c] − dist[a][b] − dist[c][d]
```

At a fixed moment only one of them is a valid 2-opt reconnection. The other one would be valid if the relative direction of the two removed edges were reversed, for example after a reversal of the segment between them. That is why every pair of edges stores **both** variants:

```
{a, b, c, d}   removes (a,b) and (c,d), adds (a,c) and (b,d)
{a, b, d, c}   removes (a,b) and (d,c), adds (a,d) and (b,c)
```

The second variant is written with its own forward endpoints `(d,c)`, so it is a separate move in the list. Exactly one of the two is applicable for a given orientation, and the other one may become applicable later. Both variants are also generated for the initial solution, as the task requires.



The figure shows the same four endpoints with the two possible reconnections. Only the variant whose removed edges are traversed in the stored direction is applied; the other one stays in `LM`.

2.4 Applicability: the three cases

When a move is popped from `LM`, it is tested against the current tour before anything is changed. The task distinguishes three situations:

Situation	Test	Action
A removed edge no longer exists	at least one of the stored edges is not adjacent in the current tour	remove the move from `LM`
The removed edges exist, but in the opposite relative direction	the edges are adjacent, but one is forward and the other reversed	leave the move in `LM`, do not apply it, browse the rest of `LM`
The removed edges exist in the same relative direction (both forward or both reversed)	both edges are adjacent in the stored orientation	apply the move, remove it from `LM`

For a 2-opt move the test uses the positions `pa, pb, pc, pd` of its four nodes:

```
forwardAB = (pb = pa + 1 mod M)
reverseAB = (pa = pb + 1 mod M)
forwardCD = (pd = pc + 1 mod M)
reverseCD = (pc = pd + 1 mod M)
```

```
if not (forwardAB or reverseAB) or not (forwardCD or reverseCD):
    drop the move
else if (forwardAB and forwardCD) or (reverseAB and reverseCD):
    apply the move
else:
    keep the move, skip it for now
```

For an inter-route move the test is simpler:

```
i = pos[b]                              # position of the old node
if i = -1 or pos[d] != -1:
    drop the move                       # b left the tour, or d entered it
pa = path[(i-1+M) mod M]
na = path[(i+1) mod M]
if {pa, na} != {a, c} in either order:
    drop the move                       # a removed edge no longer exists
apply the replacement
```

There is no "different relative direction" case for the node replacement: the two neighbours are unordered in the delta, so the move is applied whenever both removed edges still exist.

2.5 Generating new moves

After a move is applied, only the part of the neighbourhood that touches the changed edges can have new deltas. The implementation regenerates exactly that part and leaves the rest of `LM` alone.

Initial solution. Every inter-route position and every pair of edges is evaluated. For each edge pair both 2-opt variants are generated. This is the same 14,850-move scan as the full steepest search.

After an inter-route move. The selected node at position `i` changed from `b` to `d`. The positions whose neighbours changed are `i-1`, `i` and `i+1` (modulo `M`). The procedure is:

```
applyInter(i, d):
    old ← path[i]
    path[i] ← d

    for k in {i-1, i, i+1}:
        addInter(k)                     # replace path[k] by every unselected
node

    for q not in {i-1, i, i+1}:
        evaluate replacing path[q] by old
                                        # old is now unselected and can re-
enter anywhere

    for f in 0 .. M-1:
        addEdges(i-1, f)                # the two new edges around position i
        addEdges(i, f)
```

The separate loop over `q` is needed because the removed node `old` is now unselected and can be inserted at every other position. The three `addInter` calls cover only the positions whose neighbours changed.

After a 2-opt move. The segment `path[i+1 .. j]` is reversed. The nodes at positions `i`, `i+1`, `j` and `j+1` change neighbours, and the two new boundary edges are at positions `i` and `j`:

```
applyEdge(i, j):                    # reverse path[i+1 .. j]
    reverse the segment
    update pos for the reversed nodes

    for k in {i, i+1, j, j+1}:
        addInter(k)

    for f in 0 .. M-1:
        addEdges(i, f)
        addEdges(j, f)
```

All other adjacent pairs of nodes are unchanged as **undirected** pairs. Their stored deltas remain exact; a reversal only flips the relative direction of some pairs, and the other orientation variant was already stored when the pair was created. This is why no full re-scan is necessary.

2.6 The LM-driven steepest search

The search follows the alternative algorithm from the assignment:

```
SteepestLM(path):                   # path = RandomSolution(seed)
    x  ← copy(path)
    LM ← empty min-heap

    for i ← 0 to M-1:
        addInter(i)
        for j ← i+2 to M-1:
            addEdges(i, j)

    repeat:
        skipped ← empty list
        applied ← false

        while LM is not empty:
            m ← LM.popMin()
            result ← check(m)
            if result = missing edge:
                continue                # m is removed from LM
            if result = wrong direction:
                skipped.append(m)       # keep m, browse further
                continue
            x ← apply(m)                # m is removed from LM
            applied ← true
            break

        if not applied:
```

```
            break                           # no applicable improving move left

        for m in skipped:
            LM.push(m)                      # deltas are still exact
    until false
```

The moves skipped in one pass are pushed back after the applied move. A skipped move is not lost: its removed edges still exist, only their relative direction is wrong at the moment, so the stored delta is still exact.

The implementation does **not** re-evaluate the whole neighbourhood when `LM` becomes empty. This is not needed: by the rule in section 2.5, every applicable improving move is either already in `LM` or was generated after the last accepted move. An empty `LM` therefore means that the current solution is a local optimum for the full neighbourhood. Section 4.5 checks this on all 800 baseline and LM runs (200 per method per instance).

2.7 The baseline

The reference is the steepest edge-exchange search from Task 3 with a random start, `RandomStartSteepestEdges`. It uses the same starting solution, the same inter-route exchanges, the same 2-opt exchanges and the same delta formulas, but it recomputes the whole neighbourhood in every iteration and keeps no list. It is the "without the mechanism" baseline required by the task.

The two Task 4 candidate searches are also registered in the same run. They are not part of the LM comparison; they are included only so the timings and objectives come from one process. The LM mechanism is implemented separately from candidate moves, as required.

2.8 Why the list is faster

One full steepest iteration evaluates

```
M(N−M) + M(M−3)/2 = 100·100 + 100·97/2 = 14,850 moves
```

on these instances: 10,000 inter-route exchanges and 4,850 2-opt exchanges. The LM search pays that cost once, for the initial solution. After an accepted inter-route move it evaluates at most

```
3(N−M) + (M−3) + 4M  ≈  3·100 + 97 + 400  ≈  800 moves
```

and after an accepted 2-opt move at most

```
4(N−M) + 4M  =  4·100 + 400  ≈  800 moves.
```

The `4M` term counts both orientation variants of every new edge pair. The per-iteration evaluation count therefore drops by about a factor of `14,850 / 800 ≈ 18`. The measured wall-clock speedup is smaller

(3.5–3.8×). The initial scan is unchanged, every heap push and pop costs more than a simple delta loop, and the search may pop many moves that turn out to be non-applicable or missing.

2.9 Ties and equivalence with the full search

Both searches always apply a move with the smallest available delta. The full scan breaks a tie by keeping the first move in its fixed iteration order; the heap breaks it by heap order. When several moves have the same delta, the two searches can accept different moves and then continue on a different path. This is the only difference between the two methods; the neighbourhood and the acceptance rule are the same.

The experiment shows this: the two methods end with identical tours in 174 of 200 TSPA runs and 175 of 200 TSPB runs, and both always end in a full local optimum. The average objective difference is below 0.04% on both instances (section 4).

3. Experiment setup

- `N = 200`, `M = 100`, two instances.
- **200 runs per method per instance.** Run `r` starts from `RandomSolution` with seed `r`; every method uses the same 200 starts.
- 5 methods × 2 instances × 200 runs = **2000 solutions**: `RandomSolution`, the full steepest search, the two Task 4 candidate searches and the LM search.
- The LM search is deterministic for a fixed start. The only randomness is the starting solution; the list order among equal deltas is deterministic.
- Times are measured around the strategy call only. They come from one run of the program and vary between runs; only comparisons inside the same run are meaningful.
- The two Task 4 candidate searches and the full search are registered in the same benchmark run as the LM search, so their timings are directly comparable.
- Candidate lists are built once per instance before the timed loop, as in Task 4; they are not charged to the first run. The LM search has no preparation step.
- The solution checker from the course (`Solution checker.xlsx`) validates every run.

To reproduce:

```
go run . -repeat 200 -json results_task5.json
python3 tools/check_with_excel.py results_task5.json \
    --out report/excel_check_task5.json
python3 tools/make_report_task5.py results_task5.json --local-optima \
    --excel-check report/excel_check_task5.json
```

The default `go run .` registers the two baselines, the two candidate searches and the LM search. To run only the two required methods and `RandomSolution`, pass an empty candidate list:

```
go run . -candidates "" -repeat 200 -json results_task5_lm_only.json
```

`make_report_task5.py` writes the Task 5 figures to `report/figures_task5/` and prints the tables below.

4. Results

4.1 Objective values

Method	TSPA.csv av (min–max)	TSPB.csv av (min–max)
RandomSolution	261,353.1 (238,111 – 291,409)	207,261.0 (186,229 – 225,881)
RandomStartSteepestEdges	76,168.9 (73,580 – 80,490)	52,662.4 (50,328 – 55,056)
RandomStartSteepestEdges10Candidates	80,218.4 (75,324 – 85,156)	52,723.0 (50,441 – 55,895)
RandomStartSteepestEdges13Candidates	78,345.8 (75,325 – 83,113)	52,600.8 (50,394 – 55,062)
RandomStartSteepestEdgesLM	76,154.2 (73,580 – 80,490)	52,679.9 (50,328 – 55,056)

The full-neighbourhood row reproduces Report 4 exactly (76,168.9 and 52,662.4), which confirms that the baseline is the same algorithm. The LM search gives almost the same results:

- **TSPA:** 76,154.2 against 76,168.9, a difference of **−14.7** (−0.02%).
- **TSPB:** 52,679.9 against 52,662.4, a difference of **+17.5** (+0.03%).
- The min–max ranges are **identical** on both instances.
- The best solutions are identical: 73,580 on TSPA (run 171) and 50,328 on TSPB (run 101).

The two candidate rows are previous methods and are shown for context. Their behaviour is the same as in Report 4: on TSPA they are clearly worse than the full neighbourhood (about +4,050 and +2,177), while on TSPB the 13-candidate search is slightly better on average (−61.6) and the 10-candidate search slightly worse (+60.6).

4.2 Running times

Method	TSPA.csv av (min–max)	TSPB.csv av (min–max)
RandomSolution	0.01 ms (0.01 – 0.59)	0.01 ms (0.01 – 0.07)
RandomStartSteepestEdges	8.05 ms (6.89 – 9.85)	8.03 ms (7.00 – 9.57)
RandomStartSteepestEdges10Candidates	1.48 ms (1.31 – 3.29)	1.52 ms (1.31 – 3.23)
RandomStartSteepestEdges13Candidates	1.97 ms (1.73 – 3.74)	1.98 ms (1.73 – 3.75)
RandomStartSteepestEdgesLM	2.30 ms (1.57 – 3.35)	2.14 ms (1.46 – 3.26)

The LM search is consistently faster than the full search:

Instance	Aggregate speedup	Median per-run speedup	Min – max per-run speedup	LM faster
TSPA.csv	3.49×	3.48×	2.46× – 5.51×	200/200
TSPB.csv	3.75×	3.88×	2.46× – 5.83×	200/200

The LM search is faster in **all 200 pairs** on both instances, so the speedup is not caused by a few runs. The aggregate ratio (ratio of the average times) and the median per-run ratio differ slightly because a run with a

very small denominator has a large ratio; the aggregate number is the more conservative one.

The candidate searches are even faster (1.48–1.98 ms per run), but on TSPA they pay for this with a 2.9–5.3% worse objective. The LM search is the only method in this run that keeps the full-neighbourhood quality and is also much faster.

Times vary between runs and between processes. The Task 4 report measured the same full search at 8.31–8.34 ms in its extended run; here it is 8.03–8.05 ms. Only comparisons inside the same run should be read as speedups. All numbers in this report come from one run that registered all five methods together.

4.3 Paired comparisons

Every pair uses the same seed, so the difference is caused only by the two search mechanisms.

Instance	Comparison	Mean difference	Min – max difference	Better / worse / tie	Identical tours
TSPA.csv	LM – full	-14.7	-1,499 – +1,044	14 / 12 / 174	174/200
TSPB.csv	LM – full	+17.5	-1,163 – +1,373	9 / 14 / 177	175/200

On TSPA the LM result is better in 14 runs, worse in 12 and equal in 174. On TSPB it is better in 9, worse in 14 and equal in 177. Almost all of the "equal" runs have exactly the same tour; 174/200 TSPA and 175/200 TSPB tours are identical. The two TSPB runs that tie on the objective but have a different tour are still full local optima.

The differences are not a systematic change in quality. They appear when several moves have the same delta and the heap and the full scan pick different ones (section 2.9). The largest single differences are 1,499 on TSPA and 1,373 on TSPB, but these are individual runs; the average differences are 14.7 and 17.5, which is 0.02–0.03% of the objective.

4.4 Objective components

Component averages are rounded, so they may not add exactly to the objective.

Instance	Method	Avg tour length	Avg node cost	Avg objective
TSPA.csv	RandomSolution	164,360.7	96,992.4	261,353.1
TSPA.csv	RandomStartSteepestEdges	23,040.2	53,128.7	76,168.9
TSPA.csv	RandomStartSteepestEdges10Candidates	23,663.5	56,554.8	80,218.4
TSPA.csv	RandomStartSteepestEdges13Candidates	23,358.0	54,987.9	78,345.8
TSPA.csv	RandomStartSteepestEdgesLM	23,035.0	53,119.2	76,154.2
TSPB.csv	RandomSolution	155,274.3	51,986.7	207,261.0
TSPB.csv	RandomStartSteepestEdges	20,847.6	31,814.8	52,662.4
TSPB.csv	RandomStartSteepestEdges10Candidates	21,074.1	31,648.9	52,723.0
TSPB.csv	RandomStartSteepestEdges13Candidates	20,884.4	31,716.4	52,600.8

Instance	Method	Avg tour length	Avg node cost	Avg objective
TSPB.csv	RandomStartSteepestEdgesLM	20,856.1	31,823.8	52,679.9

The LM components are very close to the full search: on TSPA the tour is 5.2 shorter and the nodes cost 9.5 less; on TSPB the tour is 8.5 longer and the nodes cost 9.0 more. There is no clear shift towards cheaper nodes or a shorter tour, which is what we expect if the search is the same and only faster.

The candidate methods on TSPA lose on both components, as in Report 4: with `k = 10` the tour is 623 longer and the nodes cost 3,426 more; with `k = 13` the tour is 318 longer and the nodes cost 1,859 more. The LM mechanism does not have this problem because it searches the full neighbourhood.

4.5 Full-neighbourhood local-optimum check

For every baseline and LM run, a separate checker computed the best delta over the **full** neighbourhood (all inter-route exchanges and all 2-opt exchanges) and checked that it is non-negative. This is stronger than only checking that the search stopped: it shows that no improving move was missed.

Instance	Method	Runs checked	Full local optima
TSPA.csv	RandomStartSteepestEdges	200	200/200
TSPA.csv	RandomStartSteepestEdgesLM	200	200/200
TSPB.csv	RandomStartSteepestEdges	200	200/200
TSPB.csv	RandomStartSteepestEdgesLM	200	200/200

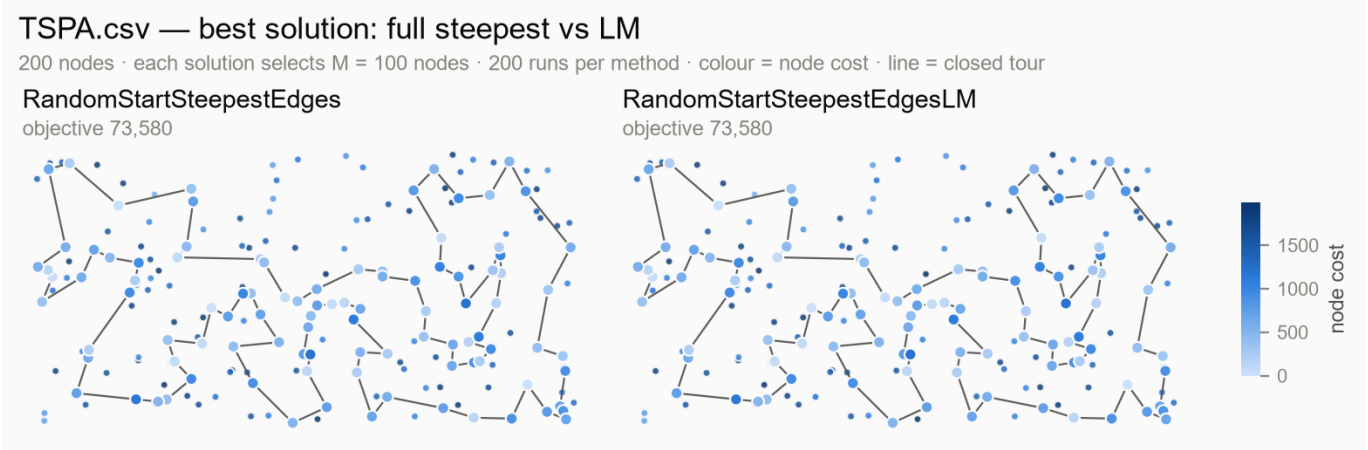
All 800 searches end in a full local optimum. In particular, the LM search never stops early because its list became empty while an improving move still existed. This confirms the rule from section 2.6 on the actual runs.

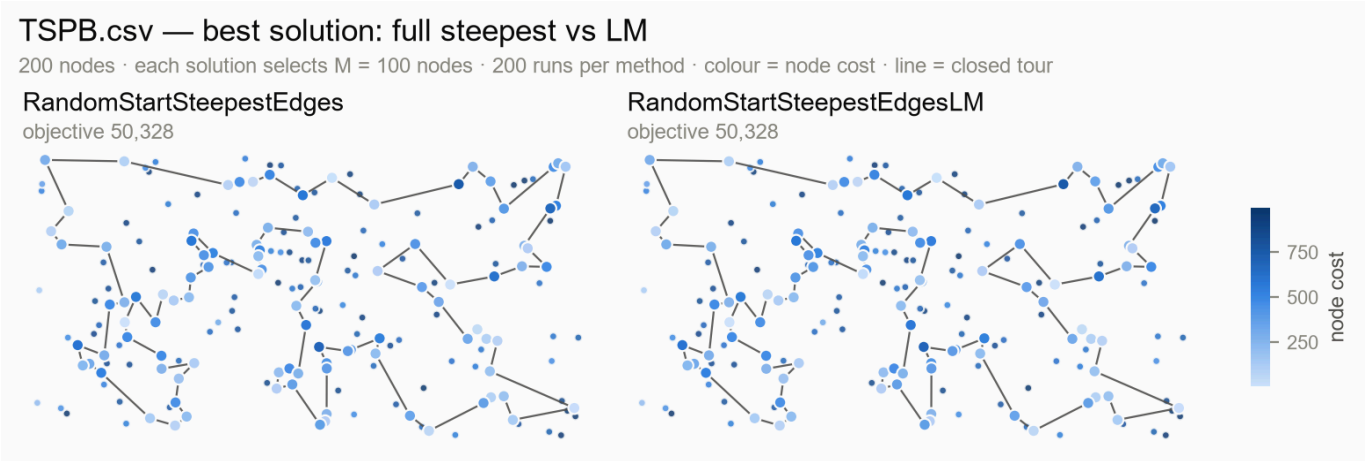
A second, stronger check compared the two searches move by move. `task5/trace_check_test.go` runs the full scan and the LM search from all 200 starts per instance and records the delta of every accepted move and the resulting tour. While the two searches are on the same tour, they always accept the **same** best delta: 19,125 accepted moves were compared on TSPA and 18,243 on TSPB before the first divergence. Every divergence happened at a tie — the two moves had the same delta but a different tour — which is exactly the behaviour described in section 2.9. No case was found where the LM search missed an improving move or used a stale delta.

```
go test ./task5
```

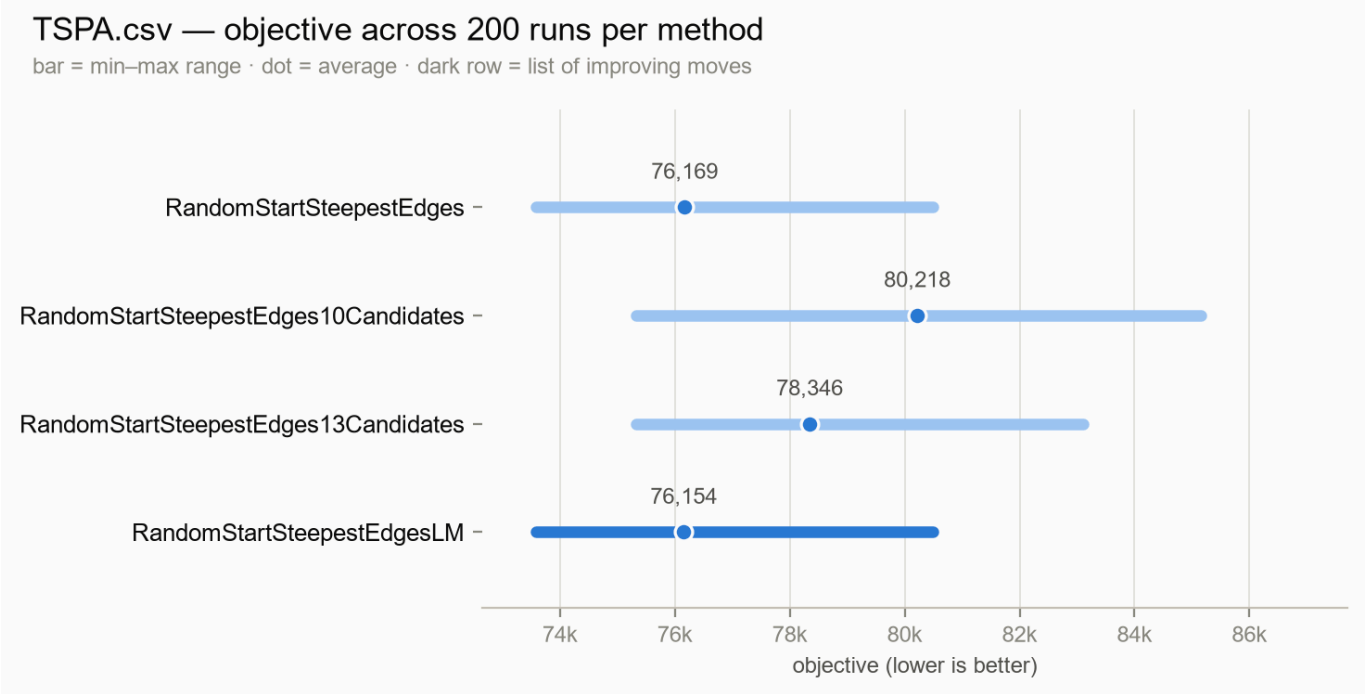
4.6 Pictures of the best solutions

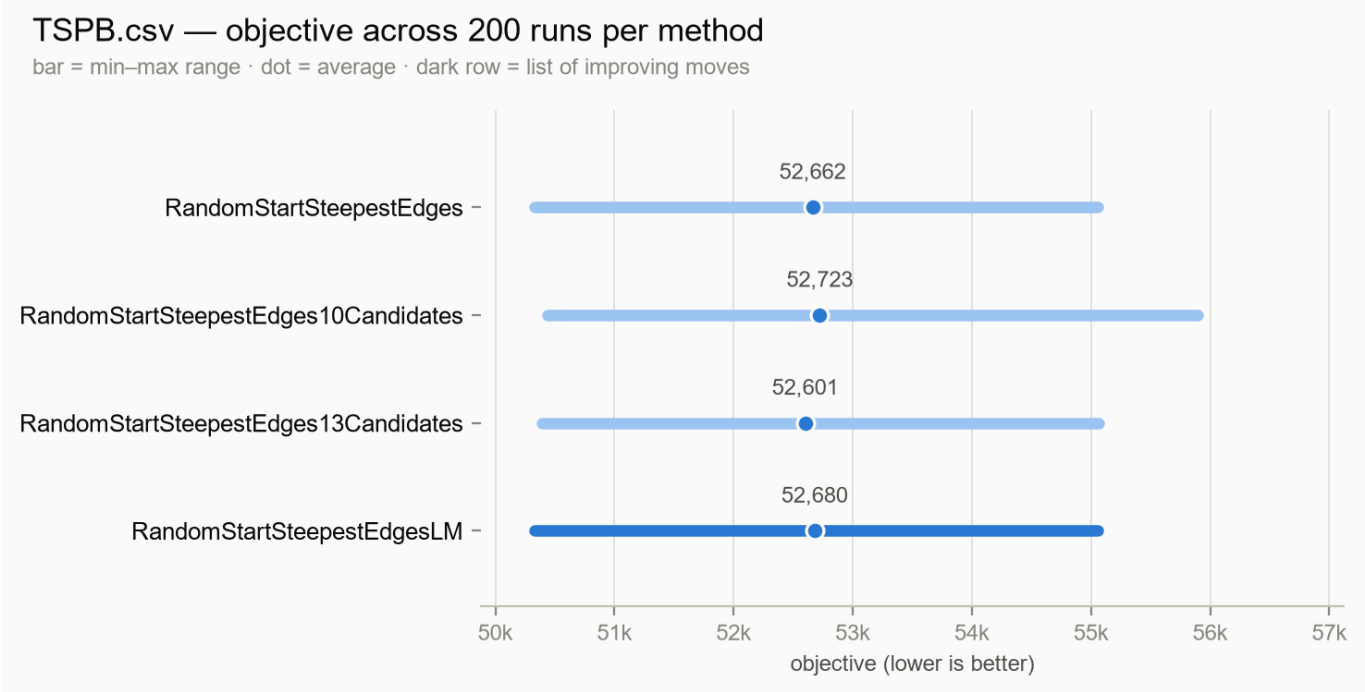
Node colour is the node cost (light blue cheap, dark blue expensive) and the dark line is the closed tour. The two panels are the full-neighbourhood search and the LM search. Their best tours are identical, so the panels look the same.



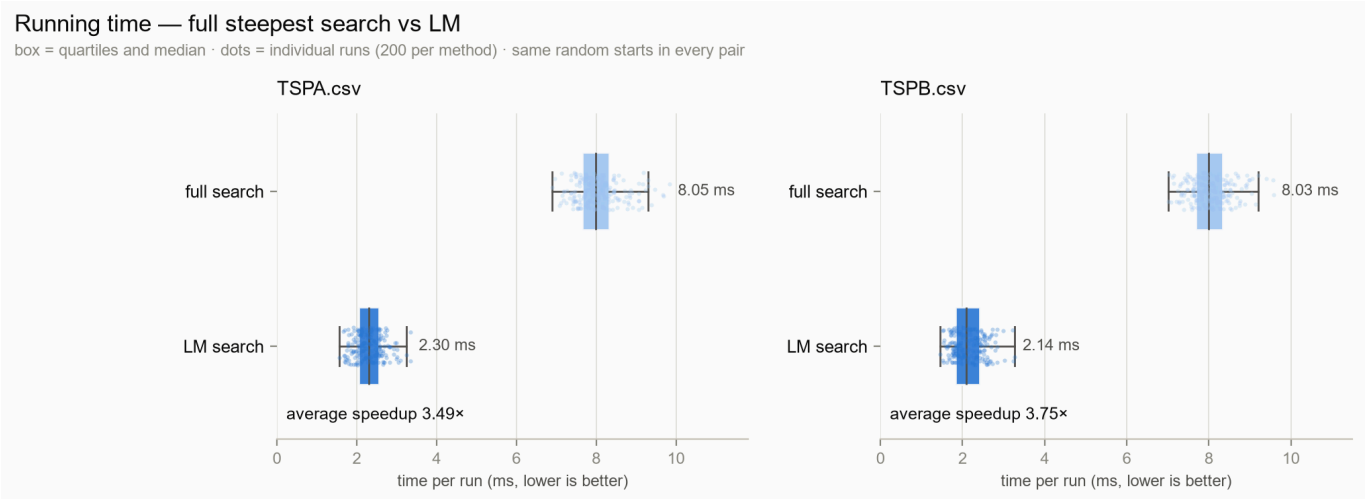


The objective ranges of the four local-search methods. The two candidate methods are included for context; on TSPA their ranges are visibly wider. The LM row is drawn in the darker accent colour.





The running-time distributions. Each dot is one run; the box shows the quartiles and the median. The full search and the LM search use the same random starts in every pair.



The time figure shows the speedup directly: the LM boxes sit at about 2.1–2.3 ms, the full search boxes at about 8.0 ms, and almost every LM dot is to the left of every full-search dot. Because the ranges overlap a little at the tails, the per-run speedup varies from run to run.

4.7 Best runs

Instance	Method	Best objective	Run	Tour length	Node cost
TSPA.csv	RandomSolution	238,111	154	146,623	91,488
TSPA.csv	RandomStartSteepestEdges	73,580	171	21,726	51,854
TSPA.csv	RandomStartSteepestEdges10Candidates	75,324	70	22,896	52,428
TSPA.csv	RandomStartSteepestEdges13Candidates	75,325	15	22,694	52,631
TSPA.csv	RandomStartSteepestEdgesLM	73,580	171	21,726	51,854

Instance	Method	Best objective	Run	Tour length	Node cost
TSPB.csv	RandomSolution	186,229	165	132,442	53,787
TSPB.csv	RandomStartSteepestEdges	50,328	101	20,812	29,516
TSPB.csv	RandomStartSteepestEdges10Candidates	50,441	149	21,229	29,212
TSPB.csv	RandomStartSteepestEdges13Candidates	50,394	149	20,711	29,683
TSPB.csv	RandomStartSteepestEdgesLM	50,328	101	20,812	29,516

The best LM run is the same solution as the best full-search run on both instances. The run column identifies the random start. Every run with its tour, objective, tour length, node cost and run index is written to `results_task5.json`; `go run .` prints the best tour of each method.

5. Solution checker

All 2000 runs were checked with the course-provided **`Solution checker.xlsx`** through `tools/check_with_excel.py`. The workbook writes the tour into `F3:F102`, uses `LOOKUP` to fetch the coordinates and costs, computes the rounded edges in `J3:J102` (`F103 = F3` closes the cycle), and reports the node cost in `I2`, the tour length in `J2` and the objective in `K2`. The script only checks that the tour has exactly `M = 100` distinct in-range ids and reads those cells back from a temporary copy, so the provided workbook is not modified.

Instance	Method	Runs checked	Feasible	Objective matches	Components match
TSPA.csv	RandomSolution	200	200/200	200/200	200/200
TSPA.csv	RandomStartSteepestEdges	200	200/200	200/200	200/200
TSPA.csv	RandomStartSteepestEdges10Candidates	200	200/200	200/200	200/200
TSPA.csv	RandomStartSteepestEdges13Candidates	200	200/200	200/200	200/200
TSPA.csv	RandomStartSteepestEdgesLM	200	200/200	200/200	200/200
TSPB.csv	RandomSolution	200	200/200	200/200	200/200
TSPB.csv	RandomStartSteepestEdges	200	200/200	200/200	200/200
TSPB.csv	RandomStartSteepestEdges10Candidates	200	200/200	200/200	200/200
TSPB.csv	RandomStartSteepestEdges13Candidates	200	200/200	200/200	200/200
TSPB.csv	RandomStartSteepestEdgesLM	200	200/200	200/200	200/200

2000 runs checked, 0 failures. All objective arithmetic in the table comes from the workbook, not from the Go solver. The additional local-optimum check in section 4.5 covers the two relevant methods separately and also passes 800/800.

6. Source code

File	Contents
`solver/problem.go`	loading the instance, the distance matrix, `TourLength` / `NodesCost` / `Objective`
`solver/solver.go`	the `Strategy` interface, the optional `Preparer` interface and the benchmark loop
`task1/strategies.go`	`RandomSolution`, used for the starting solutions and as the scale baseline
`task3/task3.go`	the full-neighbourhood steepest search, used as the baseline
`task4/task4.go`, `task4/strategies.go`	the Task 4 candidate lists and candidate steepest search, used for context
`task5/task5.go`	move representation, the min-heap, `addInter` / `addEdges`, `tryApply`, `applyInter` / `applyEdge`, `improvePathLM`
`task5/strategies.go`	`SteepestLM`
`task5/trace_check_test.go`	the move-by-move delta comparison used in section 4.5
`main.go`	command line, the `-candidates` option, the results table and the JSON export; registers the LM search
`tools/check_with_excel.py`	drives `Solution checker.xlsx`
`tools/make_report_task5.py`	Task 5 figures and tables, the paired comparison and the local-optimum check
`tools/oeviz.py`	plotting helpers

The whole mechanism is one type, `SteepestLM`, and one state struct. The state keeps the current path, the `pos` array (position or `-1`), the heap and a small scratch list for the skipped moves. `addInter` and `addEdges` are the only places where a delta is computed, and the same functions are used for the initial scan and for the updates after a move, so the delta formula is written only once.

7. Conclusions

1. **The list of improving moves gives a large speedup.** The LM search is 3.49× faster on TSPA and 3.75× faster on TSPB than the full steepest search, and it is faster in all 200 paired runs on both instances. The median per-run speedup is 3.48× and 3.88×.
2. **The quality is the same.** The average objective differs by −14.7 on TSPA and +17.5 on TSPB (0.02–0.03%), the min–max ranges are identical, the best solutions are identical, and 174/200 TSPA and 175/200 TSPB final tours are exactly the same. The remaining differences come from tie-breaking among equal deltas, not from a different search objective.
3. **The result is a full local optimum.** All 200 baseline and all 200 LM runs on each instance (800 runs in total) pass the independent full-neighbourhood local-optimum check. The update after a move covers all new moves, so an empty `LM` really means that no inter-route or 2-opt move improves the tour.
4. **The applicability rule is necessary.** A 2-opt move whose removed edges exist in the wrong relative direction must be kept, not dropped, because a later reversal can make it valid. Storing both

orientation variants for every new edge pair is what makes these kept moves usable and keeps the list complete.

5. **The speedup is limited by the initial scan and the heap.** The number of evaluations per accepted move drops by about 18×, but the end-to-end speedup is 3.5–3.8×: the initial full scan is unchanged, and heap pushes, pops and non-applicable moves cost more than a simple delta loop. A hybrid with candidate lists could reduce the initial scan and the generation cost, but that is the optional combination and was not part of this experiment.
6. **What to keep.** The LM mechanism can be added to the steepest local search without changing anything else: keep the same neighbourhood, store moves by node ids, keep both 2-opt variants, apply the three-case check, and regenerate only the moves around the changed edges. In this experiment this gives the same quality at about one third of the time on both instances.